Progressive Education Society'sModern College of Engineering, Pune-05.(An Autonomous Institute Affiliated to Savitribai Phule Pune University) Department of S.Y. B. Tech [AIDS]

Assignment No: 11

SUBJECT: Java Programming

Name: Om Mane

Class: SY BTech (AIDS)

Section: A

Roll Number: 26040

Title / Problem Statement :

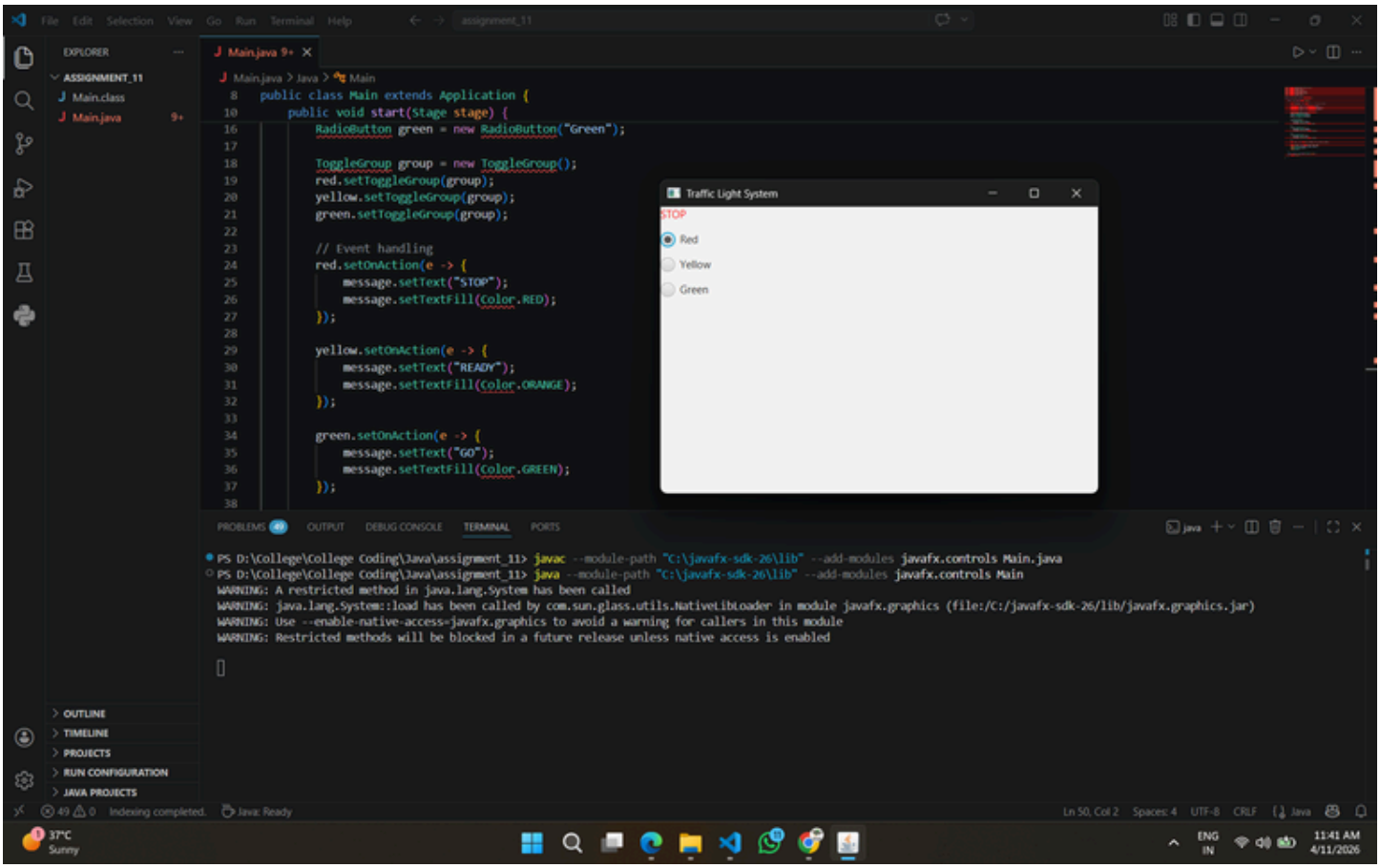
Develop JavaFX application that simulates a traffic light system. The GUI lets the user select one of three lights: red, yellow or green with radio buttons. on selecting a button an appropriate message with "stop" or "ready" or "go" should appear above the buttons in a selected color, initially there is no message shown.

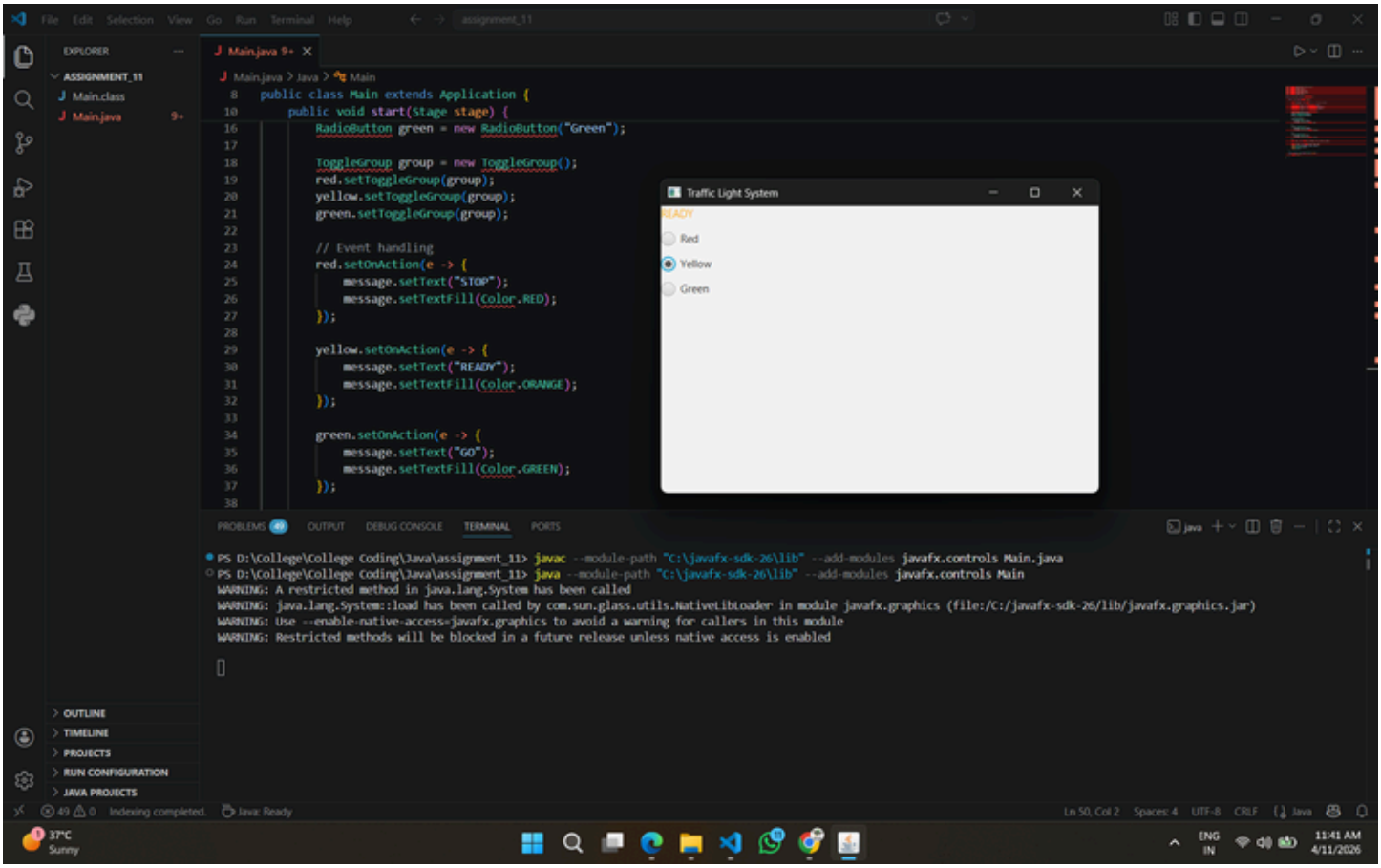
Input :

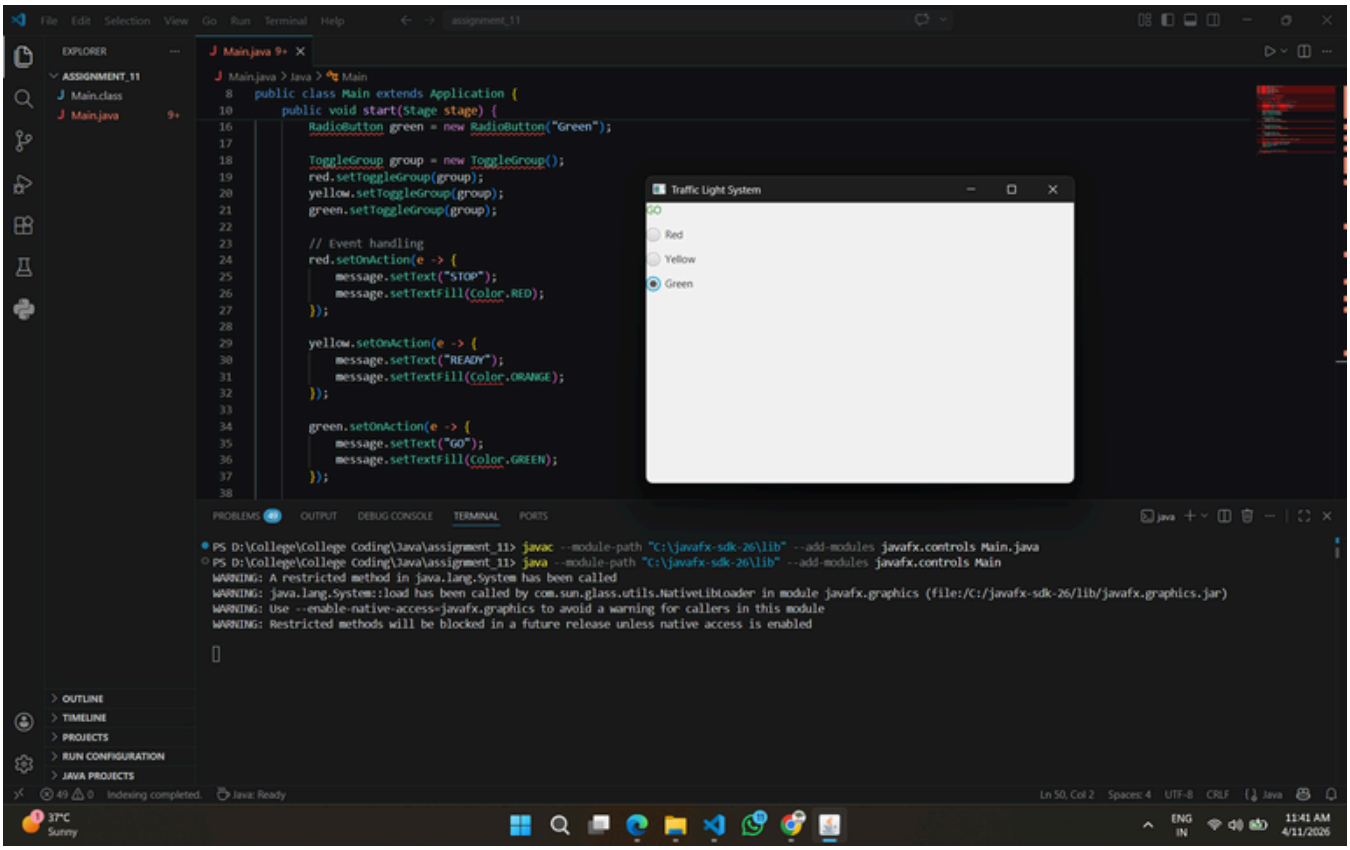
```
import javafx.application.Application;

import javafx.scene.Scene;

import javafx.scene.control.*;

import javafx.scene.layout.VBox;

import javafx.scene.paint.Color;

import javafx.stage.Stage;


public class Main extends Application {


    public void start(Stage stage) {


        Label message = new Label(""); // initially empty


        RadioButton red = new RadioButton("Red");

        RadioButton yellow = new RadioButton("Yellow");

        RadioButton green = new RadioButton("Green");


        ToggleGroup group = new ToggleGroup();

        red.setToggleGroup(group);

        yellow.setToggleGroup(group);

        green.setToggleGroup(group);


        // Event handling

        red.setOnAction(e -> {

            message.setText("STOP");

            message.setTextFill(Color.RED);

        });
```

```java
        yellow.setOnAction(e -> {
            message.setText("READY");
            message.setTextFill(Color.ORANGE);
        });

        green.setOnAction(e -> {
            message.setText("GO");
            message.setTextFill(Color.GREEN);
        });

        VBox root = new VBox(10, message, red, yellow, green);

        Scene scene = new Scene(root, 250, 200);
        stage.setTitle("Traffic Light System");
        stage.setScene(scene);
        stage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Output :


Main.java
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
STOP
Red
Yellow
Green
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled


Main.java
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
READY
Red
Yellow
Green
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled


Main.java
public class Main extends Application {
public void start(Stage stage) {
RadioButton green = new RadioButton("Green");
ToggleGroup group = new ToggleGroup();
red.setToggleGroup(group);
yellow.setToggleGroup(group);
green.setToggleGroup(group);
// Event handling
red.setOnAction(e -> {
message.setText("STOP");
message.setTextFill(Color.RED);
});
yellow.setOnAction(e -> {
message.setText("READY");
message.setTextFill(Color.ORANGE);
});
green.setOnAction(e -> {
message.setText("GO");
message.setTextFill(Color.GREEN);
});
Traffic Light System
GO
Red
Yellow
Green
PS D:\College\College Coding\Java\assignment_11> javac --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main.java
PS D:\College\College Coding\Java\assignment_11> java --module-path "C:\javafx-sdk-26\lib" --add-modules javafx.controls Main
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.glass.utils.NativeLibLoader in module javafx.graphics (file:/C:/javafx-sdk-26/lib/javafx.graphics.jar)
WARNING: Use --enable-native-access=javafx.graphics to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled